

Infrastructure Bootcamp Assignment - Week 9

SDA2026-ORJWAN

Deployment Type: AWS Web Application Deployment with ALB + ASG + S3

S3 Website Hosting

- Bucket Name: orjwan-clarusway
- Static hosting enabled
- Public access configured with bucket policy
- index.html + logo.png + sda.png uploaded
- Endpoint tested with curl showing HTTP 200 OK

Auto Scaling Group (ASG)

- Launch Template: orjwan-nginx-template
- Auto Scaling Group Name: orjwan-asg
- Desired Capacity: 2 Instances
- Health checks via EC2 and ALB
- Custom User Data used to install nginx and copy index.html from S3

Application Load Balancer (ALB)

- ALB Name: orjwan-alb
- DNS: orjwan-alb-1583318478.eu-north-1.elb.amazonaws.com
- Target Group: includes EC2 instances
- HTTP listener on port 80
- Verified round-robin with curl

Verification Commands

```
curl -I http://orjwan-clarusway.s3-website.eu-north-1.amazonaws.com
curl -s http://orjwan-alb-1583318478.eu-north-1.elb.amazonaws.com | grep hostname
for i in {1..5}; do curl -s http://orjwan-alb-1583318478.eu-north-1.elb.amazonaws.com |
grep hostname; done
```

Screenshots Included:

- S3 Bucket setup
- Static website working
- index.html in EC2
- nginx running
- Load balancer working
- curl 200 OK
- ASG showing 2 instances

General configuration

AWS Region

Europe (Stockholm) eu-north-1

Bucket type [Info](#)

☒ General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

Bucket name [Info](#)

orjwan-clarusway

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a

Copy settings from existing bucket - *optional*

Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Successfully edited static website hosting.

orjwan-clarusway [Info](#)

[Objects](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [Access Points](#)

Bucket overview

AWS Region
Europe (Stockholm) eu-north-1

Amazon Resource Name (ARN)
[arn:aws:s3::orjwan-clarusway](#)

Creation date
April 30, 2025, 11:32:11 (UTC+05:00)

orjwan-clarusway [Info](#)

[Objects](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [Access Points](#)

Objects (3)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Name	Type	Last modified	Size	Storage class
index.html	html	April 30, 2025, 11:44:37 (UTC+03:00)	2.1 KB	Standard
logo.png	png	April 30, 2025, 11:44:36 (UTC+03:00)	30.5 KB	Standard
sda.png	png	April 30, 2025, 11:44:36 (UTC+03:00)	22.5 KB	Standard

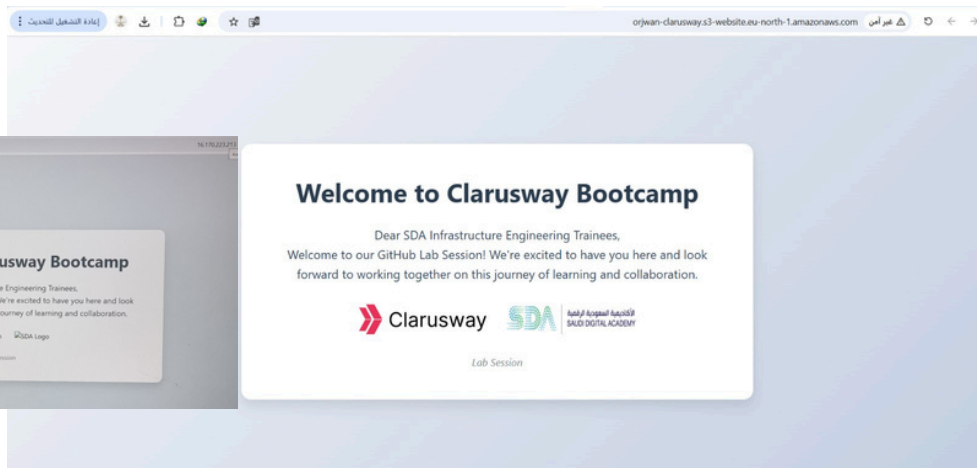
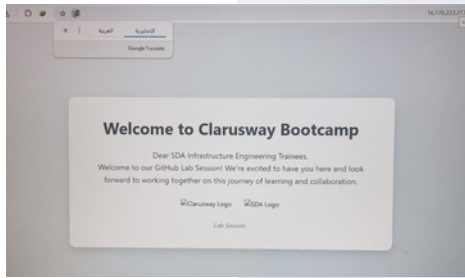
Successfully edited bucket policy.

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

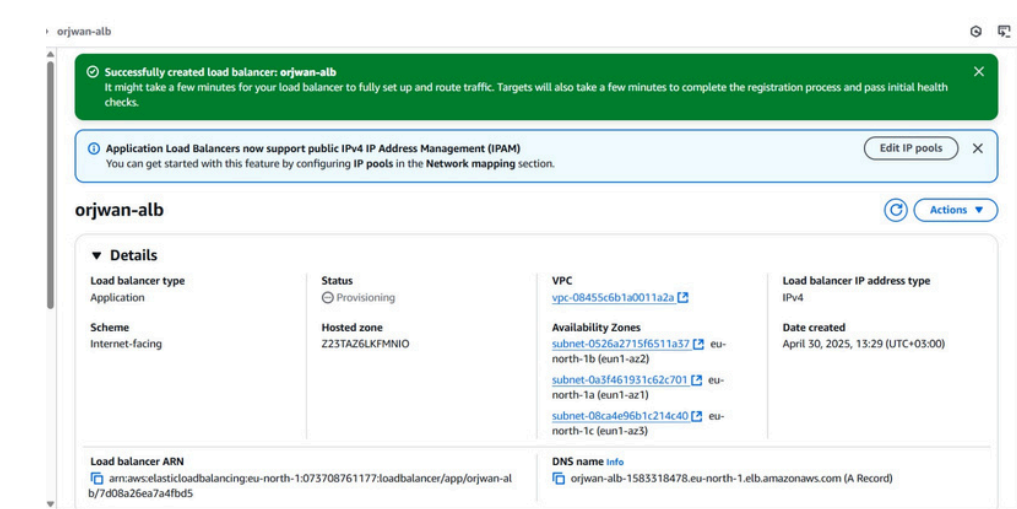
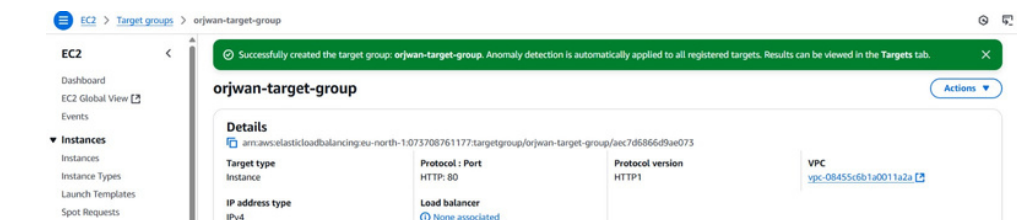
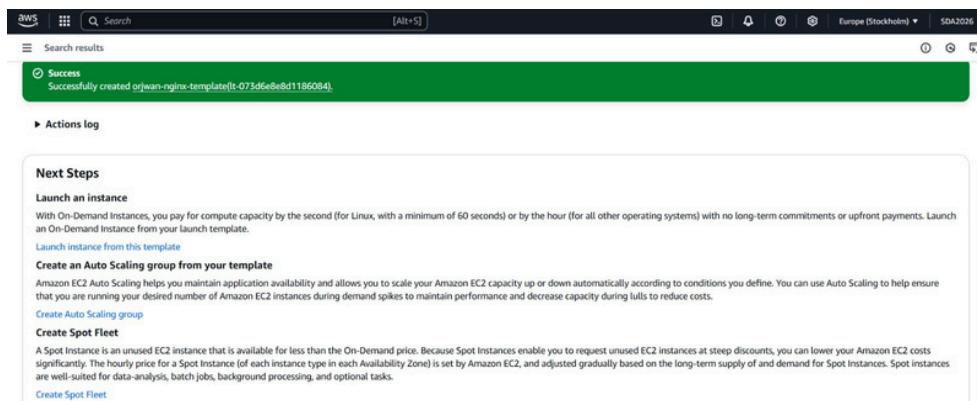
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3::orjwan-clarusway/*"
    }
  ]
}
```

[Copy](#)



```
C:\Users\Hp>curl -I http://orjwan-clarusway.s3-website.eu-north-1.amazonaws.com
HTTP/1.1 200 OK
x-amz-id-2: V8Gqq2BVKxQBkhoePwRATgUkC7iWeFX3v4m+0OheSAy9zYQLZb7sGADY+CnhvcrNtpM7iCwP4JQ=
x-amz-request-id: 0VRRQTHH84Q1BT66
Date: Wed, 30 Apr 2025 09:03:16 GMT
Last-Modified: Wed, 30 Apr 2025 08:44:37 GMT
ETag: "f13d44ec5be2837ef38b795b4c0c083f"
Content-Type: text/html
Content-Length: 2184
Server: AmazonS3

C:\Users\Hp>|
```



```
Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

last login: Wed Apr 30 11:46:53 2025 from 13.48.4.203
ec2-user@ip-172-31-26-65 ~]$ sudo aws s3 cp s3://orjwan-clarusway/index.html /usr/share/nginx/html/
fatal error: Unable to locate credentials
ec2-user@ip-172-31-26-65 ~]$ sudo aws s3 cp s3://orjwan-clarusway/index.html /usr/share/nginx/html/
download: s3://orjwan-clarusway/index.html to ../../usr/share/nginx/html/index.html
ec2-user@ip-172-31-26-65 ~]$ sudo systemctl restart nginx
ec2-user@ip-172-31-26-65 ~]$ ls -l /usr/share/nginx/html
total 16
-rw-r--r-- 1 root root 3650 Feb 11 01:59 404.html
-rw-r--r-- 1 root root 3693 Feb 11 01:59 50x.html
drwxr-xr-x 2 root root 27 Apr 30 11:50 icons
-rw-r--r-- 1 root root 2184 Apr 30 08:44 index.html
-rw-r--r-- 1 root root 368 Feb 11 01:59 nginx-logo.png
-rwxrwxrwx 1 root root 14 Feb 11 02:00 poweredby.png -> nginx-logo.png
```

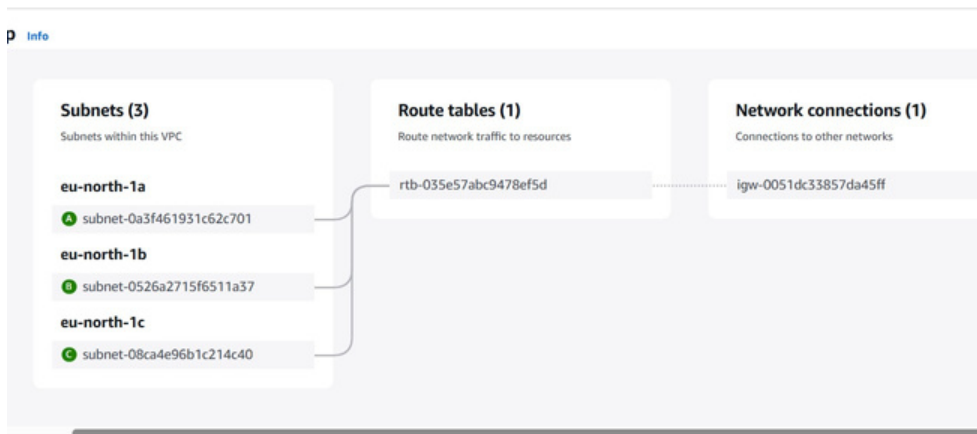
Instances (1/3) Info

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
ORJWAN	i-0374e7349438493eb	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1a	ec2-16-1...
	i-06e6a8dcfe14501a4	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1a	ec2-13-64
	i-071c638c048783977	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	ec2-13-5...

i-0374e7349438493eb (ORJWAN)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags



Listeners and rules (1) Info

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate
HTTP:80	Forward to target group - orjwan-target-group (100%) - Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable